

Exam — Databases - Summer 2026

Exam rules

- Cheat sheet: 2 double-sided A4 papers or 4 single-sided A4 papers.
- No stickies attached to the cheat sheet.
- Write your name on the paper.
- If you have a question during the exam, ask.
- Use your own A4 papers that you brought. If you need more, ask staff.

Tasks A — Explain Database Terms

Explain the following five terms in 2-3 sentences each and give one concrete database example for each term.

- Primary key
- Foreign key
- Table
- Row
- SQL query

Tasks B — Multiple Choice: True or False

For each statement, mark exactly one column. For every correct answer, you receive 2 points. For every wrong answer, 2 points are deducted. If you leave a row empty or mark both columns, you receive 0 points for that row.

Statement	True	False
A primary key identifies each row in a table.		
A foreign key can reference a primary key in another table.		
A table row is also called a record.		
The SELECT statement is used to delete rows from a table.		
The COUNT(*) function counts rows.		
A WHERE clause filters rows before they are returned.		
Every table must have a foreign key.		

Statement	True	False
<i>NULL means the same as the number 0.</i>		

Tasks C – Normalization and SQL Constraints

A small library stores book loans in one table:

```
1 CREATE TABLE library_loans (  
2     loan_id INT,  
3     loan_date DATE,  
4     due_date DATE,  
5     member_id INT,  
6     member_name VARCHAR(100),  
7     member_email VARCHAR(100),  
8     book_id INT,  
9     book_title VARCHAR(150),  
10    author_name VARCHAR(100),  
11    genre VARCHAR(50),  
12    return_date DATE,  
13    PRIMARY KEY (loan_id, book_id)  
14 );
```

1. Identify three functional dependencies that are visible in the table. Use the form 6 P.

1 A -> B, C

(A describes B and C)

2. Name two concrete anomalies or redundancies that can occur in this table. For each one, refer to specific columns from the schema. 6 P.
3. Decompose the table into relations in 3NF. Write relation schemas with primary keys underlined using (PK) and foreign keys marked with (FK). You do not need SQL syntax in this subtask. 14 P.
4. Write SQL CREATE TABLE statements for two of your relations. Each statement must include a primary key and at least one useful constraint. At least one statement must include a foreign key. 8 P.

Tasks D – Query Writing

Use the following relational schema:

```
1 Member(member_id, name, email)
2 Book(book_id, title, genre)
3 Loan(loan_id, member_id, book_id, loan_date, return_date)
```

SQL

Primary keys are the first listed attributes. *Loan.member_id* references *Member.member_id*.
Loan.book_id references *Book.book_id*.

1. Write a query that returns the names and emails of members who have at least one open loan. A loan is open if *return_date* IS NULL. 6 P.
2. Write a query that returns the total number of loans per book genre. Return genre and *loan_count*. 8 P.
3. Write a query that returns all members who have never borrowed a book. 8 P.

Tasks E — Optimized Design Task: Course Registration System

Design a database for the following restricted course registration scenario.

A university has departments. Each department has a unique name and offers courses. Each course belongs to exactly one department and has a course code. Lecturers work for exactly one department. A course can be offered multiple times, for example in different semesters. Each course offering has a semester, a year, and exactly one lecturer. Students can enroll in course offerings. An enrollment has an enrollment date, a status, and an optional grade. A student has a name, email address, and matriculation number. Email addresses and matriculation numbers are unique when known, but email addresses may be missing.

Use surrogate integer IDs where useful. You may add attributes if needed, but the required information above must be represented.

1. Draw or describe the ER model. Include the required entity types, relationships, and cardinalities. Write cardinalities explicitly if your drawing is ambiguous. 12 P.

Grading checklist: entities (4 P), required attributes and identifiers (3 P), cardinalities (3 P), handling repeated course offerings and enrollments (2 P).

2. Transform your ER model into a relational model. Write every relation as Table(attribute, attribute, ...), mark primary keys with (PK) and foreign keys with (FK). 14 P.

Grading checklist: correct tables (4 P), primary keys (3 P), foreign keys (4 P), optional values and uniqueness constraints (2 P), no unnecessary duplicated department/lecturer/student data (1 P).

3. Write PostgreSQL CREATE TABLE statements for CourseOffering and Enrollment. Include primary keys, foreign keys, and constraints for required values where appropriate. 10 P.

Grading checklist: syntactically valid DDL (2 P), PKs (2 P), FKs (3 P), NOT NULL/optional values handled correctly (2 P), useful status or year constraint (1 P).

4. Write a SQL query that returns all currently enrolled students with their course title and lecturer name. A student is currently enrolled if status = 'active'. 8 P.

Grading checklist: correct joins (4 P), filter for current enrollments (2 P), selected output columns and aliases are understandable (2 P).

In total, you can achieve 100 + 0 points. You have achieved ____ P. of 100 points.

Points	100–90	89–80	79–70	69–60	59–51	50–0
Value	A	B	C	D	E	F

Suggestion for the Solution – Exam

Task	Achieved Points
Tasks A – Explain Database Terms	____ / 0
Primary key: attribute or attribute set that uniquely identifies each row in a table. Example: student_id in Student(student_id, name).	____ / 2
Foreign key: attribute that references a key in another table and connects related rows. Example: Enrollment.student_id references Student.student_id.	____ / 2
Table: collection of rows with the same columns. Example: a Book table stores one row per book.	____ / 2
Row: one record in a table. Example: one row in Customer contains one customer's id, name, and email.	____ / 2
SQL query: statement used to ask the database for data or change data. Example: SELECT name FROM Student; returns student names.	____ / 2
Tasks B – Multiple Choice: True or False	____ / 0
True	____ / 2
True	____ / 2
True	____ / 2
False	____ / 2
True	____ / 2
True	____ / 2
False	____ / 2
False	____ / 2
Tasks C – Normalization and SQL Constraints	____ / 34
Accept any three correct dependencies, for example: member_id -> member_name, member_email; book_id -> book_title, author_name, genre; loan_id -> loan_date, due_date, member_id; (loan_id, book_id) -> return_date.	____ / 6

Award up to 3 points per well-explained anomaly. Examples: member names/emails are repeated for every borrowed book; book title/author/genre are repeated for every loan; changing a member email requires many updates; deleting the last loan of a book can lose book information; a new book or member cannot be stored before a loan if using this table only.

____ / 6

Expected structure: Member(member_id PK, member_name, member_email), Book(book_id PK, book_title, author_name, genre), Loan(loan_id PK, loan_date, due_date, member_id FK), and LoanItem(loan_id PK/FK, book_id PK/FK, return_date). Award points for correct keys, FKs, decomposition, and removal of partial dependencies.

____ / 14

Award points for syntactically plausible SQL, primary keys, foreign key use, and relevant constraints such as NOT NULL, UNIQUE(member_email), or due_date >= loan_date.

____ / 8

Tasks D – Query Writing

____ / 22

Example: SELECT DISTINCT m.name, m.email FROM Member m JOIN Loan l ON l.member_id = m.member_id WHERE l.return_date IS NULL;

____ / 6

Example: SELECT b.genre, COUNT(*) AS loan_count FROM Book b JOIN Loan l ON l.book_id = b.book_id GROUP BY b.genre;

____ / 8

Example: SELECT m.* FROM Member m LEFT JOIN Loan l ON l.member_id = m.member_id WHERE l.loan_id IS NULL; Alternative with NOT EXISTS is also correct.

____ / 8

Tasks E – Optimized Design Task: Course Registration System

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Expected entities: Department, Lecturer, Course, CourseOffering, Student, Enrollment. Relationships: Department 1:n Lecturer; Department 1:n Course; Course 1:n CourseOffering; Lecturer 1:n CourseOffering; Student n:m CourseOffering through Enrollment. Award checklist points as specified.

____ / 12

Expected model: Department(department_id PK, name UNIQUE), Lecturer(lecturer_id PK, department_id FK, name, email UNIQUE), Course(course_id PK, department_id FK, course_code UNIQUE, title, credits), CourseOffering(offering_id PK, course_id FK, lecturer_id FK, semester, year), Student(student_id PK, name, email UNIQUE NULL, matriculation_no UNIQUE), Enrollment(student_id PK/FK, offering_id PK/FK, enrollment_date, status, grade). Equivalent normalized designs are acceptable.

____ / 14

CourseOffering DDL should reference Course and Lecturer and require semester/ year. Enrollment DDL should reference Student and CourseOffering, use a primary key, require enrollment date/status, and allow grade to be nullable. PostgreSQL syntax should be plausible.	_____ / 10
Example: SELECT s.name AS student_name, c.title AS course_title, l.name AS lecturer_name FROM Enrollment e JOIN Student s ON s.student_id = e.student_id JOIN CourseOffering co ON co.offering_id = e.offering_id JOIN Course c ON c.course_id = co.course_id JOIN Lecturer l ON l.lecturer_id = co.lecturer_id WHERE e.status = 'active';	_____ / 8
	_____ / 100 + 0 P.